Alireza Montajab

Electrical Engineering | Digital Electronics

Personal Information

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Education

2023–2027 **Bachelor of Science in Electrical Engineering**, *Shahid Beheshti University, Tehran, Iran*
GPA: 16.50/20 (3.30/4)
Specialisation: Digital Electronics – expected graduation June 2027

2023 **High School Diploma**, *Allameh Tabatabai High School, Tehran, Iran*
GPA: 4.0/4.0

Selected Courses

CMOS Integrated Circuit Design, *Grade: 20/20 (4/4)*
Power System Analysis I, *Grade: 19.9/20 (4/4)*
Digital Signal Processing, *Grade: 19.25/20 (4/4)*
Fundamentals of Mechatronics, *Grade: 17.5/20 (4/4)*
Digital Systems Laboratory I, *Grade: 17.5/20 (4/4)*
Digital Systems II, *Grade: 17/20 (4/4)*

Awards and Honors

- Awarded a **Full Tuition Scholarship** at Shahid Beheshti University (2023–2027), granted on national entrance examination rank.
- Ranked within the **Top 1%** of the Iranian University Entrance Exam (Konkur), 2023.

Projects

Industry Projects

Spring 2026 **Handheld Barcode Terminal**, *Tensor Intelligent Machines*
- Built a handheld terminal that reads a commercial USB barcode scanner through the USB Host HID stack.
- Using ESP32-S3, ESP-IDF and FreeRTOS, with a custom carrier PCB designed in Altium Designer.

Winter 2025 **RFID Reader Board**, *Tensor Intelligent Machines*
- Designed a surface-mount RFID reader board with a load switch, transistor drivers and a four-button interface.
- Using Altium Designer, with a fully sourced bill of materials.

Winter 2025 **Ultrasonic Ranging Module**, *Tensor Intelligent Machines*
- Measured distance with an HC-SR04 sensor at microsecond resolution using hardware timer capture.
- Using an STM32G0 microcontroller, with timeouts for missing echoes.

Academic Projects

Spring 2026 **Magnetic Levitation System with Visual Feedback**, *Fundamentals of Mechatronics, Supervisor: Dr. Nouri-Manzar*
- Held a steel ball under an electromagnet, using an ESP32-CAM as the position sensor instead of IR or Hall-effect sensors.
- Ran the image processing and the PID controller on the same microcontroller.

Fall 2025 **Water Level and Temperature Control System**, *Digital Systems II, Supervisor: Dr. Asharioun*
- Controlled the water level and temperature of a tank with two control loops on an STM32.
- Bare-metal firmware without sensor libraries, with timer-capture ultrasonic sensing and an LCD and keypad menu.

Fall 2025 **Wi-Fi Controlled Smart Lighting**, *Digital Systems II*
- Built an ESP32 web server with REST endpoints that controls four lighting channels.

Fall 2025 **Transmission Line Parameter Calculator**, *Power System Analysis I*
- Wrote a MATLAB tool that computes the inductance and capacitance of bundled three-phase lines.

Spring 2025 **8-Bit Arithmetic Datapath**, *Digital Systems I, Supervisor: Dr. Farhad Pouladi*
- Designed an 8-bit datapath from discrete 74LS logic: adders, multiplexers, a decoder and shift registers.
- Simulated and tested in Proteus.

Spring 2025 **H-Bridge Motor Driver PCB**, *Electronics II*
- Designed a four-transistor bidirectional DC motor driver.
- Using Altium Designer.

Spring 2024 **Op-Amp RLC Network Analysis**, *Electrical Circuits I*
- Derived the transient and steady-state response of an op-amp RLC network.
- Verified the results in LTspice.

Research Interests

- Embedded Systems and Real-Time Computing
- Digital and VLSI Design
- Control Systems and Robotics
- Signal Processing and Sensing

Experiences

Teaching Experience

2024 **Altium Designer PCB Design Workshop**, *Instructor*, Mechatronics Student Association, Shahid Beheshti University
Taught schematic design, component libraries, board layout and design-rule checking.

2024 **Electrical Circuits**, *Teaching Assistant*, Shahid Beheshti University
Held weekly problem-solving sessions and graded assignments.

Work Experience

2026–Present **Embedded Hardware Engineer**, *Tensor Intelligent Machines*, Tehran, Iran
Firmware and PCB design for robotic systems. Diagnosis and repair of faulty circuit boards on malfunctioning robots.

2025–2026 **Engineering Intern**, *Tensor Intelligent Machines*, Tehran, Iran
Embedded firmware and PCB design; converted to a full-time engineering role.

Volunteer

- Organising committee member, 22nd International ISC Conference on Information Security and Cryptology (ISCISC), Shahid Beheshti University.
- Organising committee member, 15th International Conference on e-Commerce (ECDC), Tehran.
- Member, Electrical Engineering Scientific Association, Shahid Beheshti University.
- Conducted and published an alumni interview for the IEEE Student Branch newsletter, Shahid Beheshti University.

Skills

Languages:

English IELTS Academic: Overall 7.0
(Listening: 8.0, Reading: 7.5, Speaking: 7.0, Writing: 6.5)

Software and Tools:

C, C++, Embedded C, MATLAB, FreeRTOS

STM32 (HAL, CubeMX), ESP32 (ESP-IDF), AVR, Arduino

Altium Designer, LTspice, Proteus, CodeVisionAVR, Fritzing